

ANEES HUSSAIN

EXCELLENCE IN EDUCATION SINCE 1989



DAWOOD 1

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ANEE'S HUSSAIN

DARWAZA

ENGLISH

Choose the correct SYNONYM for the following

- 1) Ameliorate
a) Understand
✓ b) Improve

- ✓ c) Eliminate
d) Confined

- 2) Abysmal
✓ a) Bottomless
c) Fatalist

- b) Addition
d) Valuable

- 3) Blandishment
a) Punishment
c) Unwilling

- b) Cheating
✓ d) Flattery

- 4) Brochure
a) News
✓ c) Pamphlet

- b) Magazine
d) Annals

- 5) Circumvent
✓ a) Caution
d) Diversity

- b) Frustrate
d) Quote

Choose the correct ANTONYM for the following

- 6) Acrimonious
✓ a) Bitter
c) Plain

- ✓ b) Soothing
d) Intelligent

- 7) Archetype
a) Copy
c) Origin

- b) Generous
d) Factory

- 8) Belligerent
a) Loud
c) Peaceful

- b) Interested
d) Popular

- 9) Bustle
a) Ado
✓ c) Stir

- b) Rustle
d) Quiet

- 10) Cajole
a) Synthesize
c) Coax

- b) Antagonize
d) Proper

Point out which underlined part of the following sentence is incorrect.

- 11) There have been heavy rainfall yesterday.
A B C D

- 12) Aftab as well as his class fellows are playing hockey.
A B C D

- 13) Neither the head constable nor other policemen is injured.
A B C D

- 14) Every leaf and every flower proclaim the glory of God.
A B C D

- 15) Six miles are a long distance.
A B C D

Fill in the following blanks with the most appropriate word.

- 16) _____ should not hate the poor.

Belli -> war/conflict

Anthrop -> human

Amphi / Amphib ->

Dual / both

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- ☒ a) The rich
☐ b) Rich
☐ c) A rich
☐ d) An rich
- 17) We take _____ at 8 am daily.
☐ a) the breakfast
☒ b) A breakfast
☐ c) breakfast
☐ d) none of these
- 18) The more you work, _____ it is.
☐ a) better
☒ b) the better
☐ c) a better
☐ d) more better
- 19) _____ Himalayas lie in the north of Pakistan.
☐ a) A
☐ b) The
☐ c) An
☐ d) none
- 20) Today we need _____ to advocate for us.
☐ a) the Quaid-e-Azam
☐ b) Quaid-e-Azam
☒ c) a Quaid-e-Azam
☐ d) none of these

Read the following comprehensions and answer the questions that follow

PASSAGE 1

In Asia and much of the Third World, trees are still destroyed in the old-fashioned way: they are cut down for fuel and cropland. In Europe, there is new and potentially more deadly culprit. The Germans call it "Waldsterben", the dying forest syndrome. But the disease is far more than a German phenomenon. Since it was first observed by German scientists in the autumn of 1980, the mysterious malady has raced across Europe, blighting woods in countries as far apart as Sweden and Italy. Explanations for the epidemic range from a cyclic change in the environment to a baffling form of tree cancer. But the most convincing evidence points to air pollution. Indeed, saving the rapidly deteriorating forests of Europe will probably require a two-pronged strategy: an offensive campaign that includes the breeding of pollution-immune trees and a defensive scheme that calls for reductions in toxic emissions. But both will require more money than is currently being spent on such measures, as well as total commitment to protecting the environment.

- 21) According to this passage, which one of the following statements is correct?
- ☐ a) There is less damage in Asia than in Europe
☐ b) More forests are dying in Germany than anywhere else in Europe
☒ c) A cyclic change in the environment is responsible for deforestation
☒ d) Air pollution is the main culprit of destroying European forests
- 22) The dying forest syndrome is a disease that
- ☐ a) is peculiar to the forests of Asia
☒ b) has spread rapidly over the forests of Europe
☐ c) is confined to the forests of Germany
☐ d) has affected forests all over the world
- 23) The writer suggests that
- ☐ a) it is no longer possible to grow trees in industrialized areas
☒ b) pollution immune trees will absorb toxic emissions
☐ c) all pollution-prone trees should be destroyed
☒ d) it is not possible to grow trees that remain unaffected by pollution
- 24) The writer's approach toward the problem of forest devastation is one of
- ☐ a) tolerance
☐ b) indifference
☒ c) well thought-out strategy
☐ d) despondency
- 25) Saving the trees of European forests
- ☒ a) requires a much bigger budget

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DAWOOD 1

- b) appears to be a hopeless task and therefore pointless to undertake
- c) should not be difficult because of the advances in experimental research
- d) demands vigilance and punitive measures against those who cut down the trees



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Dawood 1

CHEMISTRY

1. One atm is equal to _____ Pascal
a.) 760 ☒ b.) 101325
c.) 14.7 d.) 1.01325
2. If $V_1 = 5$ liters, $P_1 = 2$ atm, $T_1 = 273^\circ\text{C}$, $T_2 = 0^\circ\text{C}$ and $V_2 = ?$ when $P_2 = 1$ atm
a.) 5 liters ☒ b.) 2.5 Liters
c.) 12.5 Liters d.) 10 Liters
3. _____ lies in U.V region
a.) Balmer Series ☒ b.) Lyman Series
c.) Paschen Series d.) All of the above
4. The maximum number of electrons that can be present in a main energy level is
a.) $2n^2$ ☒ b.) $2n$
c.) 2 d.) n
5. The two electrons in same orbital should have:
a.) Similar spin ☒ b.) opposite spin
c.) parallel spin d.) Perpendicular spin
6. Mass of electron is:
a.) 9.1×10^{-31} Kg ☒ b.) 19×10^{-31} Kg
c.) 9×10^{-30} Kg d.) 9.1×10^{-30} Kg
7. Change of Enthalpy is denoted
a.) H ☒ b.) ΔH
c.) E d.) ΔE
8. Rate constant for gaseous reaction is written as:
a.) K_c ☒ b.) K_p
c.) K_n d.) K_{sp}
9. Oxidation state of oxygen in per-oxides
a.) 2 ☒ b.) -1
c.) -2 d.) 1
10. Rate of reaction:
a.) Increases as the reaction proceeds
b.) Decreases as the reaction proceeds ☒
c.) Remain same as the reaction proceeds
d.) May increase or decrease as the reaction proceeds
11. Photochemical reaction usually have the order
a.) One ☒ b.) Two
c.) Three d.) Zero
12. _____ are biocatalysts.
a.) Organic acids ☒ b.) Organic bases

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13. Metalloids gives:
 a.) Metal
 b.) Amphoteric
 c.) Homogenous
 d.) Heterogeneous
14. Fifth period of modern periodic table contains:
 a.) 2
 b.) 6
 c.) 12
 d.) 18
15. Electro negativity difference for ionic bond must be greater than:
 a.) 1.6
 b.) 1.7
 c.) 1.8
 d.) 1.9
16. The Measurement of degree of disorder
 a.) Entropy
 b.) Enthalpy
 c.) Rate of Reaction
 d.) Temperature
17. The unit of K_c for reaction $N_2 + O_2 \rightleftharpoons 2NO$
 a.) $Mol\ dm^{-3}$
 b.) $Mol\ dm^{-1}$
 c.) $mol^2\ dm^{-3}$
 d.) None
18. If two lone pairs are present then bond angle of tetrahedral compound reduce to
 a.) 109.5°
 b.) 107.5°
 c.) 104.5°
 d.) None
19. _____ is the formula of fluorspar?
 a.) Na_3AlF_6
 b.) CaF_2
 c.) $CaF_2 \cdot 3CaF_2$
 d.) AlF_3
20. HF is:
 a.) Oxidizing agent
 b.) Reducing agent
 c.) Weak acid
 d.) Strong acid
21. _____ is called oil of vitriol.
 a.) H_2SO_4
 b.) HNO_3
 c.) HCl
 d.) CH_3COOH
22. Nobel metals are dissolved in:
 a.) Conc. H_2SO_4
 b.) Aqua Regia
 c.) Conc. HCl
 d.) All of these
23. Catenation is the ability of carbon to grow in _____
 a.) Size
 b.) Number
 c.) Line
 d.) Long Chains
24. Which is the most volatile compound?
 a.) HI
 b.) HCl
 c.) HBr
 d.) HF

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25. Di-ethyl ketone methyl n- propyl ketone are an example of:

- a.) Chain Isomerism b.) Positional Isomerism
c.) Metamerism Isomerism d.) Functional group Isomerism



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DAWOOD PHY 1

PHYSICS

1. The speed of sound wave travelling through air doesn't depend upon
☒ A. air temperature ☐ B. atm pressure
☒ C. gasses in air ☐ D. thermal conductivity
2. A 2Ω resistor is connected for 0.5 min to a P.D of 1K.V. the heat developed in resistor is
☒ A. 36 kJ ☐ B. 36 K J
☐ C. 60 kJ ☐ D. 360 KJ
3. Electric flux is the
☐ A. scalar product of the electric field and the surface area
☐ B. flow of electric lines per unit area
☐ C. scalar quantity
☒ D. all of them
4. Which is more penetrating?
☒ A. α - rays ☐ B. β - rays
☐ C. γ - rays ☒ D. X rays
5. An object is placed at 30 cm from a lens of focal length 15cm its image size will be
☐ A. 6 cm ☐ B. 15 cm
☐ C. 8 cm ☒ D. 30 cm
6. Time period of simple pendulum is
☒ A. $T = 2\pi \sqrt{\frac{l}{g}}$ ☐ B. $T = 2\pi \sqrt{\frac{l}{g}}$
☐ C. $T = 2\pi \sqrt{\frac{l}{g}}$ ☐ D. $T = 2\pi \sqrt{\frac{l}{g}}$
7. In 1932 James Chadwick discovered
☒ A. Electron ☐ B. Proton
☐ C. Neutron ☒ D. None
8. Controlled fission reaction takes place in
☒ A. Cyclotron ☐ B. Nuclear reactor
☐ C. Sun ☐ D. Earth
9. Center of Newton's Ring is:
☒ A. Dark ☐ B. Both
☐ C. Bright ☒ D. None
10. Find the value of 'a' if two vectors $A = 2i + aj + k$ and $B = 4i - 3j + 7k$ are perpendicular to each other.
☐ A. 20 ☐ B. 10
☒ C. 10 ☐ D. 20
11. The rate of change of angular momentum is equal to the
☒ A. torque ☐ B. Mass
☐ C. force ☒ D. energy
12. Find the work done on the body which starts from rest and moving when a force of 10N is applied on it for 4s if its mass is found to be 4Kg.
☐ A. 1.0 J ☐ B. 200 J
☒ C. 25J ☐ D. 300J
13. A ball is the height of 80m. Find the time taken by ball to reach ground.
☒ A. 16 sec ☐ B. 4 sec
☐ C. 32 sec ☐ D. 8 sec
14. Unit of radioactivity is.
☒ A. Kg/s ☐ B. Decay/sec
☐ C. Becquerel ☐ D. None

$$F = m \cdot \frac{dv}{dt} = m \cdot a$$

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15. Light waves are always.
 A. Longitudinal
 B. Transverse
 C. High speed
 D. High frequency
16. At the H_{max} the velocity of the projectile is:
 A. Maximum
 B. Minimum
 C. Remains same
 D. None of the above
17. The distance between two adjacent antinodes is
 A. $\frac{\lambda}{2}$
 B. $\frac{\lambda}{4}$
 C. λ
 D. 2λ
18. The wavelength of a standing wave produced in a 4m long closed pipe with 4 loops will be:
 A. 1.14m
 B. 2.4m
 C. 30m
 D. 3m
19. The area of Force time graph will yield.
 A. Momentum
 B. Acceleration
 C. Impulse
 D. Power
20. The turn ratio of a transformer having 400 turns on primary coil and 380 turns on a secondary coil will be:
 A. 1.5
 B. 3:4
 C. 4:3
 D. 1:4
21. By covering one slit of Young's double slit experiment.
 A. Reduce intensity of fringes
 B. Reduce wavelength
 C. Reduce fringe spacing
 D. No fringes will produce
22. Which of the following unit is not a unit of energy?
 A. Electron Volt
 B. Kilo Watt Hour
 C. Wattsec
 D. Volt
23. Unit vector is used to specify what property of a vector?
 A. Magnitude of a vector
 B. Direction of a vector
 C. Dimension of a vector
 D. None of the above
24. Free fall motion is an example of:
 A. Uniform velocity
 B. Variable velocity
 C. Uniform Acceleration
 D. None of the above
25. The rate of change of angular momentum is called
 A. Force
 B. Centripetal force
 C. Torque
 D. Centripetal acceleration

18

$$L = \frac{\lambda}{4} + \frac{\lambda}{4} + \frac{\lambda}{4} + \frac{\lambda}{4}$$

$$\frac{\lambda + \lambda + \lambda + \lambda}{4}$$

$$L = \frac{4\lambda}{4} \quad \lambda = \frac{4L}{4} = \frac{4(4)}{4} = \frac{16}{4} = 4$$

MATHEMATICS

- If $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ then $A^{-1} =$
 - $\begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$
 - $\begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix}$
 - $\begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix}$
 - None
- If $\begin{bmatrix} 2 & 3 \\ 1 & x \end{bmatrix} \begin{bmatrix} 1 & y \\ x & -1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 2 & 3 \end{bmatrix}$ then value of $x + y = ?$
 - 0
 - 1
 - 2
 - None of these
- If $P(x_1, y_1)$ is inside the circle then?
 - $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c < 0$
 - $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c > 0$
 - $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$
 - None of these
- If $f(x) = 2x - 1$ then $f(2) = ?$
 - 3
 - 3
 - 5
 - 5
- $d/dx (e^x) = ?$
 - $5e^x$
 - e^x
 - e^x
 - e^x
- The area of quadrilateral (0, 0), (5, -3), (5, 0) and (5, -5)
 - 20 sq. units
 - 10 sq. units
 - 10 sq. units
 - None
- If $y = 2x^2 + 3x - 4x + 1$ then $y'' = ?$
 - $120x^2 + 18$
 - $120x^2 - 18$
 - $4x + 1$
 - All of these
- If the line has slope m then its equation in the form of slope intercept form while intercept is c
 - $y = mx + c$
 - $y = m + c$
 - $y = mx$
 - All of these
- The solution set of the equation $3x^2 - 7x - 2 = 0$
 - $\frac{-\sqrt{63} \pm \sqrt{63}}{6}$
 - $\frac{-\sqrt{63} \pm \sqrt{63}}{6}$
 - $\frac{-\sqrt{63} \pm \sqrt{63}}{6}$
 - $\frac{-\sqrt{63} \pm \sqrt{63}}{6}$
- Direction cosine of vector (3, 7, -4)
 - $\left(\frac{3}{\sqrt{84}}, \frac{7}{\sqrt{84}}, \frac{-4}{\sqrt{84}}\right)$
 - $\left(\frac{3}{\sqrt{84}}, \frac{7}{\sqrt{84}}, \frac{-4}{\sqrt{84}}\right)$
 - $\left(\frac{3}{\sqrt{84}}, \frac{7}{\sqrt{84}}, \frac{-4}{\sqrt{84}}\right)$
 - $\left(\frac{3}{\sqrt{84}}, \frac{7}{\sqrt{84}}, \frac{-4}{\sqrt{84}}\right)$
- The period of $\tan \frac{1}{2}x$ is
 - 5π
 - $5/\pi$
 - $5\pi/2$
 - None of these
- $\sqrt{-b}$
 - $\sqrt{b}$
 - $\sqrt{bi}$
 - $\sqrt{ib}$
 - $\sqrt{bi}$
- 210°
 - $2\pi/3$
 - $\pi/6$
 - $7\pi/6$
 - $\pi/3$
- Integral of $\sqrt{x}$ w.r.t. x ?
 - $\frac{1}{2}\sqrt{x} + c$
 - $\frac{1}{2}$
 - $\frac{1}{\sqrt{x}} + c$
 - $\frac{1}{2\sqrt{x}} + c$
- The n th term of the sequence $2\sqrt{2}, 8, 16\sqrt{2}, \dots$ is
 - $(2\sqrt{2})^n$
 - $(2\sqrt{2})^{n+1}$
 - $(2\sqrt{2})^{n-1}$
 - $(\sqrt{2})^n$

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16. 5 boys are running in 100m race, 8 boys are running for long jump and 1 boy is running in 100m race. In how many ways prize can be distributed?
- a. 370
b. 280
c. 380
d. None of these
17. Two dice were thrown simultaneously. Find the probability of having sum equals 13
- a. 1
b. 0
c. 1/12
d. 1/6
18. The constant term in the expansion of $(2x + \frac{1}{x})^8$
- a. 1120
b. 70
c. 140
d. All of these
19. If $U = \{1, 2, 3\}$, $A = \{1, 2\}$, $B = \{2, 3\}$ and $C = \{1, 3\}$ then $(A \cap B) \cap C = ?$
- a. $\{1, 2\}$
b. $\{2, 3\}$
c. $\emptyset$
d. $\{1, 2, 3\}$
20. $\sin 2\theta$
- a. $\frac{2 \tan \theta}{1 + \tan^2 \theta}$
b. $\frac{2 \tan \theta}{1 - \tan^2 \theta}$
c. $\frac{2 \tan \theta}{1 + \tan^2 \theta}$
d. $\frac{2 \tan \theta}{1 - \tan^2 \theta}$
21. The equation of ellipse is
- a. $x^2 + y^2 + 2x + 2y + 1 = 0$
b. $x^2 + y^2 + 2x + 2y + 2 = 0$
c. $x^2 + y^2 + 2x + 2y + 3 = 0$
d. $x^2 + y^2 + 2x + 2y + 4 = 0$
22. The sum of two numbers is 25 and sum of their square is 125, find the difference
- a. 12
b. 16
c. 14
d. 0
23. If $\cos \theta = \frac{1}{2}$ and $\sin \theta = \frac{\sqrt{3}}{2}$ while $0^\circ < \theta < 360^\circ$?
- a. 60°
b. 240°
c. 45°
d. 300°
24. The general term of sequence 3, 7, 11, ... is
- a. $4n - 1$
b. $5n - 1$
c. $4n + 1$
d. $5n + 1$
25. If $f(x) = 2$ and $g(x) = x + 4$ then $\log_2(x) =$
- a. 4
b. 2
c. 1
d. 3

STUDENT
GROUP:
ENGL

1
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Choose the correct

- 1) Crest Fallen
a) Key
c) Aim
- 2) Deplore
a) Gay
c) Regret
- 3) Dissonance
a) Entertain
c) Discord
- 4) Emissary
a) Whimsy
c) Hamlet
- 5) Empathy
a) Purgatory
c) Fear

Choose the

- 6) Consequence
a) Obstacle
c) Discrepancy
- 7) Deficient
a) Shortage
c) Excess
- 8) Deviate
a) Depart
c) Adhere
- 9) Exacerbate
a) Intensify
c) Alleviate
- 10) Eminent
a) Prominent
c) Obscure

Pos

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DAWOOD 2

ENGLISH

Choose the correct SYNONYM for the following

- 1) Crest Fallen
a) Key
c) Aim
b) Annoy
☒ d) Dejected
- 2) Deplore
a) Gay
☒ c) Regret
b) Glee
d) Praise
- 3) Dissonance
a) Entertainment
☒ c) Discord
b) To Defect
d) Thesis
- 4) Emissary
a) Whimsical Notion
c) Hammer Flat
b) Traditional Beliefs
☒ d) Secret Agent
- 5) Empathize
a) Purgation
c) Fear
☒ b) Sympathize
d) Pity

Choose the correct ANTONYM for the following

- 6) Consent
a) Obey
c) Disgrace
b) Hate
☒ d) Retire
- 7) Deficit
a) Surplus
c) End
b) Limited
d) Finish
- 8) Deviate
a) Diverge
☒ c) Resemble
b) Pervade
d) Stray
- 9) Evasion
a) Response
c) Excuse
b) Pretext
d) Patent
- 10) Encroach
a) Invade
c) Infringe
b) Intrude
☒ d) Desist

Point out which underlined part of the following sentence is incorrect.

- 11) playing the harmonium and singing is difficult.
A B C D
- 12) The team are out to win the match.
A B C D
- 13) The cluster of grapes were plucked by the child.
A B C D
- 14) A box of apples is are in the car.
A B C D
- 15) The jury is are arguing among themselves.
A B C D

Fill in the following blanks with the most appropriate word.

- 16) No doubt, he has achieved much, I can't give him credit _____ all that he boasts.
a) for, about
b) an, about

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- c) for, to d) about, the
- 17) By wearing contact lenses that corrected his colour blindness, he was _____ for the job.
 a) was able for qualifying ☒ was able to qualify
 c) was ably qualified d) was ably qualifying
- 18) These terms are very harsh; I doubt the CEO will agree _____ them.
 a) for ☒ with
 c) of d) to
- 19) I'm afraid that I have to agree _____ your opinion about the new actuary.
 a) with ☒ over
 c) from d) of
- 20) _____ only chance you are left with is to re-appear for _____ paper.
 a) X, the ☒ The, the
 c) The, X d) The, a

PASSAGE

For fourteen and half months I lived in my little cell or room in Dehradun jail, and I began to feel as if I was almost a part of it. I was familiar with every bit of it, I knew every mark and dent on the whitewashed walls and on the uneven floors and on the ceiling with the moth-eaten rafters. In the little yard outside I greeted little tufts of grass and odd bits of stone as old friends. I was not alone in my cell, for several colonies of wasp and hornets lived there, and many lizards found the home behind the rafters, emerging in the evening in the search of prey.

- 21) Which of the following explains best the sentence in the passage "I was almost a part of it"?
 a) I was not alone in the cell
 b) I was familiar with every bit of the cell
 c) I greeted little tufts of grass like old friends
 d) I felt quite at home in the cell
- 22) The author of the passage seems to suggest that
 a) It is possible to adjust one-self to uncongenial surroundings.
 b) The conditions in Indian prisons are not bad
 c) It is not difficult to spend one's time in prison
 d) There is a need to improve the conditions in our jails
- 23) The passage attempts to describe:
 a) The general conditions of the country's jail
 b) The prisoner's conscious efforts to overcome the loneliness
 c) The prisoner's capacity to notice the minute details of his surroundings
 d) The prisoner's ability to live happily with other creatures

Choose the option which is closely related to the capitalized word sentence.

- 24) Book : Cupboard
 a) Glass : Drink ☒ Wine : Bottle
 c) Waste : Jug
- 25) Circle : Square
 a) Oval : Rectangle ☒ Volume : Area
 c) Sphere : Globe

1. If the density of substance
 a. 20 cm³
 b. 40 cm³
2. 20°C is equal to how?
 a. 0.5
 b. 68
3. Which of the follow
 a. CH₃COCl
 b. MgO.NO₂
4. Internal resistance
 a. Viscosity
 b. Surface ten
5. Mass of 10 ml
 a. 0.12 kg
 b. 0.120 kg
6. Which of the
 a. Cu
 b. Na
7. Which one
 a. Amf
 b. Amn
8. Which o
 a. Ten
 b. Co
 c. Ca
 d. N
9. Poly
 a.
 b.
10. Aq
 a.
 b.
- 11.

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Dawood 2

CHEMISTRY

1. If the density of substance is 3g/cm^3 and mass is 60 then what will be the volume
 - a. 20cm^3
 - b. 40cm^3
 - c. 60cm^3
 - d. 180cm^3
2. 20°C is equal to how many $^\circ\text{F}$?
 - a. 0.9
 - b. 68
 - c. 6.7
 - d. 8
3. Which of the following pair is not covalent?
 - a. $\text{CH}_4, \text{CCl}_4$
 - b. MgO, NO_2
 - c. $\text{CO}_2, \text{Al}_2\text{O}_3$
 - d. KCl, NaCl
4. Internal resistance to flow of liquid
 - a. Viscosity
 - b. Surface tension
 - c. Vapour pressure
 - d. Evaporation
5. Mass of 10 mole of carbon
 - a. 0.12 kg
 - b. 0.120 kg
 - c. 12 kg
 - d. 1.9 kg
6. Which of the following has paramagnetic properties?
 - a. Cu
 - b. Na
 - c. Ni
 - d. K
7. Which one of the compound is used as raw material for urea?
 - a. Ammonium acetate
 - b. Ammonia
 - c. Both a and b
 - d. None of these
8. Which one of the following will not effect the rate of reaction?
 - a. Temperature
 - b. Concentration
 - c. Catalyst
 - d. Material of Container
9. Polyethylene is a
 - a. Polymer
 - b. Ester
 - c. Synthetic fiber
 - d. None of these
10. Aqueous solution of NaCl on electrolysis gives
 - a. Sodium at anode
 - b. Chloride to cathode
 - c. Hydrogen to cathode
 - d. Hydrogen gas at anode
11. Hydrogen atom consist of
 - a. One electron one neutron
 - b. One electron one proton
 - c. One proton one neutron
 - d. One electron, one proton, and one neutron
12. Which of the following are isomorphous pair?
 - a. $\text{SiO}_2, \text{AgNO}_3$
 - b. $\text{CaCO}_3, \text{Na}_2\text{O}$
 - c. $\text{NaNO}_3, \text{CaCO}_3$
 - d. NaF, MgO
 - e. Both c and d
13. Maximum electron in s block are
 - a. Four
 - b. Three
 - c. Two
 - d. One

ANES HUSSAIN

Dawood 2

14. The direction and extent of chemical reaction are told by _____
- Rate
 - Equilibrium
 - Catalyst
 - None of these
15. The unit of dipole moment is _____
- Debye
 - Poles
 - Poise
 - None of these
16. Phenol forms phenoxides of the correspond metal with an _____
- Metal
 - Acid
 - Alkali
 - Carbon compound
17. Phenol is _____
- Colorless solid
 - Poisonous gas
 - Non-Poisonous gas
 - Amorphous solid
18. Hybridization of phenol is _____
- sp^3
 - sp^2
 - sp
 - None of these
19. Carbon form _____
- metallic bond
 - covalent
 - Ionic bond
 - Polar ionic bond
20. William son synthesis is used to prepared _____
- Acetone
 - Diethyl ether
 - PVC
 - Bakelite
21. Ether are isomers with _____
- Alcohol
 - Aldehyde
 - Ketones
 - None of these
22. Diethyl ether is used as _____
- Antibiotic
 - Antiseptic
 - Anesthesia
 - Analgesic
23. Marsh gas mainly contain _____
- CH_4
 - C_2H_6
 - CO
 - CO_2
24. Which has least C-C bond length _____
- Ethane
 - Ethene
 - Ethyne
 - Benzene
25. Hybridization of olefin is _____
- sp^3
 - sp^2
 - sp
 - None of these

$$12n = \frac{5}{9} T_c + T_2$$

$$1096 = \frac{5}{9} T_c + 32$$

$$1064 = \frac{5}{9} T_c$$

$$T_c = \frac{1064 \times 9}{5} = 1915.2$$

PHYSICS

1. The motion of charge particle controlled by.
 - A. electric current
 - C. electric & magnetic field
 - B. electric power
 - D. gravitational field
2. After a complete cycle of a carnot engine the internal energy
 - A. increases
 - C. remain same
 - B. decreases
 - D. none
3. Transistor is used most as
 - A. Amplifier
 - C. Switch
 - B. Moderator
 - D. All Of the Above
4. A projectile is launched for a maximum range of 400m, which of the following co-ordinates will indicate its position at 100m?
 - A. (200, 200)
 - C. (200, 100)
 - B. (100, 100)
 - D. (100, 200)
5. A force of 20N is applied on two springs connected in series combination and each spring is of 2N/m stiffness. If a mass is attached with them the deflection produce by it will be
 - A. 1m
 - C. 2m
 - B. 10m
 - D. None
6. What will be the centripetal force if the velocity is doubled and radius is increase by 4?
 - A. 4 times
 - C. Doubled
 - B. Remain same
 - D. 1/4 times
7. Precondition for lasing action in photodiode is to
 - A. Enlarge the depletion region
 - C. inversion
 - B. Achieve population
 - D. Start spontaneous emission
8. A galvanometer is converted into an ammeter by connecting with it
 - A. A low resistance in parallel
 - C. A high resistance in parallel
 - B. A low resistance in series
 - D. A high resistance in series
9. Ten resistors each of $10\ \Omega$ are connected in parallel. The net resistance is
 - A. $10\ \Omega$
 - C. $1\ \Omega$
 - B. $0.1\ \Omega$
 - D. $0.001\ \Omega$
10. 122° Fahrenheit temperature is equal to
 - A. 50°C
 - C. 323°C
 - B. 90°C
 - D. 363°C
11. What will be the decay constant of a radioactive material having half life of 1600yrs?
 - A. $2.3 \times 10^{-11}/\text{sec}$
 - C. $1.3 \times 10^{-11}/\text{sec}$
 - B. $5.3 \times 10^{-11}/\text{sec}$
 - D. None
12. A convex lens of 20 cm. focal length is used to form an erect image which is twice as large as the object. The position of object from the lens is
 - A. 5 cm
 - C. 115 cm
 - B. 10 cm
 - D. 20 cm
13. Colour of light is determined by.
 - A. Velocity in Air
 - C. Wave length
 - B. Amplitude of vibration
 - D. Frequency
14. By cutting a lens vertically its focal length of remaining pieces will:
 - A. Increases
 - C. Decrease
 - B. Remains same
 - D. None

- ANEES HUSSAIN DAWOOD PHY 2
15. Physical quantity tends to rotate a body is
 A. Force
 C. Energy
 B. Momentum
 D. Torque
16. What will be the length of the day when the radius of Earth is halved but mass remains same?
 A. 12Hrs
 C. 24Hrs
 B. 6Hrs
 D. 8Hrs
17. Transistor is used to amplify
 A. voltage
 C. both
 B. current
 D. none of the above
18. Ten equal resistors, each of resistance 10Ω is connected in parallel. The equivalent resistance of the circuit is
 A. 1Ω
 C. 100Ω
 B. 10Ω
 D. 50Ω
19. A photon is moving with speed of light, its rest mass will be
 A. Equal to electron
 C. Equal to Carbon
 B. Equal to proton
 D. Zero
20. A train of $150m$ in length is going towards north at a speed of $10m/s$. A parrot flies at a speed of $5m/s$ towards south parallel to the railway track. The time taken by the parrot to cross the train is equal to
 A. $20s$
 C. $30s$
 B. $10s$
 D. $50s$
21. Unit of static Coefficient of friction is
 A. kg/m^2
 C. dyne
 B. Newton
 D. none
22. The relative intensity (I/I_0) of the sound of a jet engine is 10^{13} . The intensity level in dB will be
 A. 13
 C. 1300
 B. 130
 D. 13
23. The point where the amplitude is zero in a standing wave pattern is called
 A. Wave number
 C. Node
 B. Anti Node
 D. None of the above
24. A bomb is dropped from an airplane moving with a horizontal velocity of 200km/h . If air resistance is neglected and stone take $5s$ to reach the ground, find the altitude of the plane:
 A. $120m$
 C. $125m$
 B. $200m$
 D. $400m$
25. The work produced when a force of $10N$ displaces an object from position $(1,2,1)$ to $(2,3,1)$ will be:
 A. $10J$
 C. $18J$
 B. $14J$
 D. $3J$

24

$$v = 0$$

$$a = 10$$

$$u = ?$$

$$t = 5 \text{ sec}$$

$$s = vit + \frac{1}{2}at^2$$

$$s = \frac{1}{2}(10)(5)^2$$

$$s = 125m$$

MATHEMATICS

(1) Let $M = \{r, s, t\}$

- i. $r \in M$
- ii. $r \subset M$
- iii. $\{r\} \in M$
- iv. $M \supset r$

- a. i only
- b. iii only
- c. i & iv

- d. i & iii
- e. All of these

(2) Two dice were rolled once, the probability of obtaining multiple of 3 is

- a. $\frac{5}{9}$
- b. 0
- c. $\frac{5}{6}$

- d. $\frac{9}{10}$
- e. None of these

(3) Which of the following function is neither odd nor even?

- a. $\{(1, 2), (4, 7), (1, 2), (0, 4), (-4, 7)\}$
- b. $\{(1, 2), (4, 7), (1, 2), (0, 0), (-4, -7)\}$
- c. $\{(x, y) : y = x^2 - 1\}$

- d. $\{(x, y) : y = x^2 - 1\}$
- e. $\{(x) : x^2 = 1\}$

(4) Cube roots of unity are

- a. $1, w, w^2$
- b. $1, w, w^2, \pm 1$
- c. $1, w, w^2, \pm i, \pm 1$

while, fourth roots are

- d. $1, w, w^2, \pm 1$
- e. None of these

(5) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n+1}\right)^n$

- a. e
- b. e^{-1}
- c. $e^{-1}e^2$

- d. $e^{-1}e$
- e. None of these

(6) Let $S = \{A, B, C, D\}$ where $A = \{a\}$, $B = \{a, b\}$, $C = \{a, b, c\}$ and $D = \phi$, then the missing elements of the composition table below are

U	A	B	C	D
A	A	B	C	A
B	B	B	C	B
C	C	C	C	C
D	A			?

- a. C, D
- b. A, A
- c. A, B

- d. B, B
- e. C, C

(7) The middle term(s) in the expansion of $(x - 2y/3)^{10}$ is/are

a. $\frac{896}{27}x^5y^5$

b. $\frac{896}{27}x^5y^4$

c. $-\frac{896}{27}x^5y^5$

d. $-\frac{896}{27}x^5y^4$

e. None of these

(8) The T_n of the sequence $1/3, 12/3, 3 \dots$ is

a. $\frac{1(4n-3)}{3}$

b. $\frac{1(4n-2)}{3}$

c. $\frac{1(4n-5)}{3}$

d. $4n-3$

e. None of these

- AMTES HUSSAIN
- (9) Any matrix of order $n \times n$ whose elements are all zero is called
- Square Matrix
 - Rectangular Matrix
 - Triangular Matrix
 - Null Matrix
 - None of these
- (10) $\sin^{-1} \frac{1}{2}$ is
- 1
 - $\frac{\pi}{6}$
 - 2
 - $-\frac{\pi}{2}$
 - 0
- (11) $\frac{d^2y}{dx^2}$ of $y = (x^3 + 2x)^2$ is
- $\frac{3x^2 + 2x}{2}$
 - $\frac{(x^2 + 2x)(3x^2 + 2)}{2}$
 - $\frac{(x^2 + 2x)(3x^2 - 1)}{2}$
 - $\frac{(3x^2 + 2x + 4x^2)}{2}$
 - Both b & d
- (12) $\int \frac{dx}{x \ln 3x}$
- $5 \ln x + C$
 - $\ln 5x + C$
 - $\ln(\ln x)^2 + C$
 - $\ln(\ln 3x) + C$
 - $\ln(\ln x) + C$
- (13) Find the area under the curve for the equation $x^2 - 3x + 6$ intersecting line $y = x + 3$
- $2\frac{1}{3}$ sq. units
 - $1\frac{1}{3}$ sq. unit
 - $1\frac{1}{3}$ sq. units
 - $3\frac{1}{3}$ sq. units
 - $2\frac{1}{3}$ sq. units
- (14) A sector cut from a circle of radius 3 cm has a perimeter of 12 cm. The area of this sector is
- 9 cm^2
 - 18 cm^2
 - 15 cm^2
 - 25 cm^2
 - None of these
- (15) Find the coordinates of the point that divides the join of A(2, -6) & B(2, 1) in the ratio 3:1 internally
- (0, 1)
 - (1, $\frac{1}{3}$)
 - (2, $-\frac{14}{3}$)
 - (1/2, 1/3)
 - (1, 1)
- (16) The center and radius of the circle having equation $2x + 6y - x^2 - y^2 = 1$ is
- (1, 3), 3
 - (3, 1), 3
 - (1, 3), 4
 - (3, 1), 4
 - (2, 3), 5
- (17) A parabola has the axis at x and passes through the point (-2, 6). Its equation will be
- $y^2 = 12x$
 - $y^2 = -12x$
 - $y^2 = 3x$
 - $y^2 = -3x^2$
 - $y^2 = 12x + 72$
- (18) The direction cosine of the vector formed from (1, -3, 2) and (3, -5, 1) are
- $\frac{2}{3}, \frac{3}{3}, \frac{-6}{3}$
 - $\frac{-2}{3}, \frac{-3}{3}, \frac{6}{3}$
 - $\frac{2}{7}, \frac{-3}{7}, \frac{-6}{7}$
 - A & B both
 - None of these
- (19) The equation of the straight line that passes through the point $(4, \sqrt{3})$ and makes an angle of $\pi/6$ radians with x-axis will be:
- $x + \sqrt{3}y + 1 = 0$
 - $x = \sqrt{3}y - 1$
 - $x + \sqrt{3}y - 1 = 0$
 - $x - \sqrt{3}y - 1 = 0$
 - $x = \sqrt{3}y$

ANES HUSSAIN

Date: _____

(20) If the distance between $(7, 1)$ and $(3, y)$ is 5 units, find the value of y

- a. 4
b. -2
c. A & B
d. -4
e. B & D

(21) If $f(x) = x^3 - 9x^2 + 15x + 3$ the relative maxima is

- a. 5
b. 1
c. 10
d. -22
e. 0

(22) Let $A = \begin{bmatrix} -2 & 0 & 3 \\ 1 & -6 & 5 \\ -1 & 2 & 0 \end{bmatrix}$, the co-factor a_{33} will be

- a. 1
b. 2
c. 3
d. 4
e. 5

(23) An AP is such that the sum of the first 8 term is 156 and sum of first 10 terms is 215. The fifth term will be

- a. 20
b. 18.5
c. 19
d. 20.5
e. 19.5

(24) If $(1 + 5i)p - 2q = 3 + 7i$ then values of p & q will be

- a. $7/5, 4/5$
b. $-7/5, 4/5$
c. $7/5, -4/5$
d. $7/5, -4/5$
e. None of these

(25) In how many ways can 4 prizes be awarded to 3 students if each student can receive at four prizes?

- a. 4!
b. 3^4
c. $3^2 - 1$
d. 4
e. 3!

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ENGLISH

Choose the correct SYNONYM for the following

- 1) Flotsam
a) Wreckage
c) A fleet
b) Fade Away
d) Line of Buoys
- 2) Forbearance
a) Patience
c) Noisy
b) Animated
d) Hysterical
- 3) Garrulous
a) Foolish
c) Playful
b) Sincere
d) Talkative
- 4) Glib
a) Fluent
c) Satisfied
b) Snide
d) Superficial
- 5) Holocaust
a) Postulated
c) Hooligan
b) Destruction
d) Grumpy

Choose the correct ANTONYM for the following

- 6) Fatal
a) Deadly
c) Beneficial
b) Dangerous
d) Baneful
- 7) Fanciful
a) Veritable
c) Capricious
b) Whimsical
d) Fantastic
- 8) Guilty
a) Noted
c) Childish
b) Childlike
d) Innocent
- 9) Gradual
a) Regular
c) Slow
b) Momentary
d) Progressive
- 10) Hypocrisy
a) Cant
c) Uprightness
b) Deceit
d) Pretense

Point out which underlined part of the following sentence is incorrect.

- 11) The woman who owns those cars live in my building.
A B C D
- 12) Each of the children are allowed one extracurricular activity.
A B C D
- 13) Everyone in my town remember the night the firehouse burned down.
A B C D
- 14) The cabinet are empowered to take this decision.
A B C D
- 15) Each of the nurses were scrupulously careful about personal cleanliness.
A B C D

Fill in the following blanks with the most appropriate word.

- 16) It is ___ interesting book. It gives ___ wonderful picture of what ___ life was like in Victoria.

OD 3

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DAWOOD 3

- a) the, a, the
b) the, the, a
c) an, a, the
d) an, the, a
- 17) Many secretaries fail because they are not
a) voracious
b) versatile
c) vindictive
d) vivacious
- 18) Ideas are more powerful than
a) legal
b) weapons
c) military
d) lethal
- 19) There are too many
a) dilettantes
b) and not enough serious workers
c) novices
d) sycophants
- 20) It is easier to
a) praise
b) than to accept criticism
c) negotiate
d) explain
- 21) The man was called hero because
a) Expressed his willingness to risk his life to save others
b) Managed to escape from fire
c) Showed great courage in fighting the fire
d) Saved a life
- 22) The man went to another room because
a) He did not know where the stairway was
b) The passage was full of smoke
c) He was extremely nervous
d) He stumbled on bundle

PASSAGE

A local man, staying on the top of the floor of an old wooden house, was awakened at midnight by fire. Losing his way in the smoke-filled passage, he missed the stairway and went into another room. He picked a bundle to protect his face from fire and immediately fell through the floor below where he managed to escape through a clear doorway. The "bundle" proved to be the baby of the Mayor's wife. The "hero" was congratulated!

- 23) Architect : Blue Print
a) Mason : Wall
b) Compass : Circle
c) High : Low
d) Glass : Jug
- 24) Argument : Debate
a) Violence : Peace
b) Fight Contest
c) Challenge : Opponent
d) Give : Take
- 25) Beef : Cow
a) Mutton : Buffalo
b) Sheep : Mutton
c) Mutton : Sheep
d) Pork : Cow

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Dawood

CHEMISTRY

- The process in which a solid directly converted to a gaseous state is called
 - Boiling
 - Evaporation
 - Sublimation
 - Freezing
- The no. of atoms of nitrogen represented in the formula NH_4NO_3 is
 - 1
 - 2
 - 3
 - 4
- Which of the following technique could be used to separate the coloured from the black ink
 - Chromatograph
 - Filtration
 - Fractional distillation
 - Precipitation
- The crystalline form of allotropes are
 - Chemically same
 - Chemically different
 - Some time same some time different
 - Depend upon substance
- Alpha rays are
 - Single negative charged particles
 - Single positive charged particles
 - Double negative charged particles
 - Double positive charged particles
- In an atom no. two electron has the same set of quantum number
 - Uncertainty principle
 - Pauli's exclusion principle
 - Hund's rule
 - Aufbau's principle
- Properties of a system, which depend upon the amount of substance present in the system are called
 - Chemical Properties
 - Intensive properties
 - Extensive properties
 - Physical properties
- The chemical formula of laughing gas is
 - N_2O
 - NO
 - N_2O_2
 - N_2O_4
- The heat required to change 1 mole of iodine solid to iodine gas is called
 - Heat of vaporization
 - Heat of sublimation
 - Heat of fusion
 - Melting point
- In the electrolysis of CuCl_2 the substance liberated at the anode is
 - Copper
 - Chloride
 - Hydrogen
 - Copper Chloride
- Factor affecting the balance of a chemical equilibrium are following except
 - Concentration
 - Temperature
 - Enthalpy
 - Pressure
- The formula for general gas equation is
 - $P_1 V_1 = P_2 V_2$
 - $V_1/T_1 = V_2/T_2$
 - $E = mc^2$
 - $P_1 V_1/T_1 = P_2 V_2/T_2$
- The electronic configuration of the noble gas is
 - $ns^2 np^5$
 - $ns^2 np^6$
 - $ns^2 np^4$
 - $ns^2 np^5$
- The electro negative of the element is increased
 - Down the group
 - Across the period from right to left
 - Across the period from left to right
 - None of the above
- Which of the following is the most suitable allotropy of sulphur
 - Plastic Sulphur
 - Monoclinic Sulphur
 - Rhombic Sulphur
 - Laughing gas
- S-block element have the following properties except

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Diamond 3

- a. Reaction with water gives hydroxides
b. Reaction with water gives different oxides
17. Melting and boiling of transition elements is high because
a. Half filled d orbital
b. Metallic bonding
18. The fat soluble vitamin is
a. Vitamin A
b. Vitamin B-12
19. The formula which indicates the relative number of different kinds of the ATOM in a molecules
a. Structural formula
b. Empirical Formula
c. Molecular formula
d. None of the above
20. Methane is obtained by which of the following method.
a. Electrolysis of sodium acetate
b. Heating sodium acetate with soda lime
c. Electrolysis of aluminium carbide
d. All of these
21. The following reaction is an example of $C_2H_4 + Cl_2 \xrightarrow{\text{heat}} C_2H_4Cl_2 + 2HCl$
a. Addition reaction
b. Reduction reaction
c. Oxidation reaction
d. Substitution reaction
22. In which of the following reaction the rate of nucleophilic substitution reaction depend upon concentration of alkyl halide only
a. SN^1
b. SN^2
c. SN^1
d. Addition reaction
23. In the phenol
a. Alcoholic group attached on the ring
b. Hydroxyl group is attached on benzene ring
c. Alcoholic group is attached on benzene ring
d. All of the above
24. PVC prepared from
a. Ethene and hydrogen chloride
b. Ethane and hydrogen chloride
c. methane and hydrogen chloride
d. acetylene and hydrogen chloride
25. commercially nitric acid is prepared by
a. oxidation of ammonia by bayer's method
b. oxidation of ammonia by Oswald process
c. Reduction of ammonia by Oswald process
d. oxidation of ammonia by hoop process

PHYSICS

1. The broadcasting a radio program a long distance, the effective technique of modulation is:
 A. FM B. PM
 C. AM D. BM
2. The color of light emitted by diode (LED) depends upon:
 A. Its Biasing voltage B. Its reverse voltage
 C. Amount of Current flowing through it D. Type of semi-conductor material
3. When number of loops in a closed pin increases then the wavelength:
 A. Increases B. Decreases
 C. Remain Constant D. None of these
4. Which quantity is decreases in Step-Down transformer?
 A. Voltage B. Power
 C. Current D. Energy
5. The S.I. unit of measuring inductance is called.
 A. OERSTED B. Gauss
 C. Weber D. Henry
6. If the potential difference across a resistor is doubled, then the rate of heat production will be.
 A. double B. Half
 C. $\sqrt{2}$ time D. Four times
7. A Anometer is used to measure:
 A. Emf B. Current
 C. Unknown Resistance D. Internal Resistance of a cell
8. The efficiency of the Carnot's engine working between 100°C and 26°C is:
 A. 20% B. 60%
 C. 40% D. None
9. The relation for conversion between centigrade and Fahrenheit temperatures is given by:
 A. $^\circ\text{F} = (9/5)^\circ\text{C} + 32$ B. $^\circ\text{C} = (9/5)^\circ\text{F} + 32$
 C. $^\circ\text{F} = (9/5)(^\circ\text{C} + 32)$ D. $^\circ\text{C} = 5/9(^\circ\text{F} + 32)$
10. The image formed by an Eye piece of compound microscope is
 A. Real and Diminished B. Real and Enlarged
 C. Virtual and Enlarged D. None of these
11. The process of emitting electron by incidenting UV light on the metal sheet is
 A. Electromagnetic B. Photo electric effect
 C. Electrophorus D. None of these
12. The center of Newton's Ring is:
 A. Bright B. Dark
 C. Both D. None
13. In Reverse Bias the diode starts behaving as a:
 A. Conductor B. Insulator
 C. Both D. None of these
14. Angular speed of daily rotation of earth in rad/hr will be:
 A. 1.3 B. 2.50

$$\omega = \frac{\theta}{t} = \frac{2\pi}{24 \times 60} \text{ rad/hr}$$

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DAWOOD PHY 3

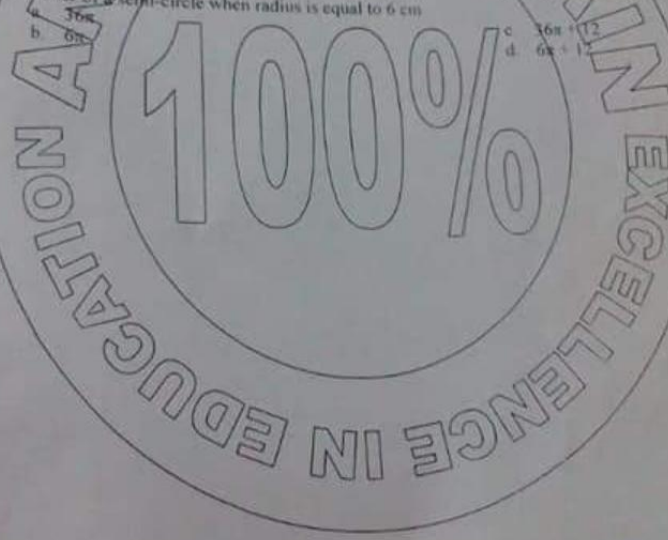
15. Unit of power of lens is _____
☒ A. Meter
☐ B. Diopter
☐ C. 0.26
☐ D. 26
16. Which is the equation of M_e ?
☐ A. $M_e = \frac{E}{R}$
☐ B. $M_e = \frac{G}{R}$
☐ C. $M_e = Gg$
☒ D. $M_e = gR^2$
17. If the force of 20N act on body at a point at axis 30° then its A_y is equal to _____
☐ A. 0
☐ B. 10
☐ C. 20
☐ D. 40
18. If the maximum height of a projectile is 40m then its maximum range will be _____
☐ A. 40m
☐ B. 80m
☐ C. 120m
☐ D. 160m
19. The unit of Angular momentum is similar with:
☐ A. Rydberg's Constant
☐ B. Planck's Constant
☐ C. Coulomb's Constant
☐ D. None
20. By adding more impurities to a semiconductor the conduction property of it will _____
☐ A. Increases
☐ B. Decreases
☐ C. Remain Constant
☐ D. None of these
21. Which quantity is remains same in Step-Down transformer?
☐ A. Voltage
☐ B. Power
☐ C. Current
☐ D. Energy
22. The colour of Black body depends on it:
☐ A. Size
☐ B. Material
☐ C. Temperature
☐ D. None
23. Which of the following is not affect by speed:
☐ A. Power
☐ B. Resistance
☐ C. Velocity
☐ D. momentum
24. which is unitless:
☐ A. Coefficient of friction
☐ B. Refractive index
☐ C. Magnification
☐ D. All of these
25. At the center of Earth the gravity is:
☐ A. Maximum
☐ B. Minimum
☒ C. Zero
☐ D. None

MATHEMATICS

1. $\lim_{n \rightarrow \infty} \frac{8n}{n+2} = ?$
 - a. 8
 - b. ∞
 - c. $\frac{8}{3}$
 - d. 3
2. Area under the curve $y = x^2 - 2x$ from $x = -1$ to $x = 3$ will be
 - a. 1
 - b. $\frac{1}{2}$
 - c. $\frac{1}{3}$
 - d. $\frac{4}{3}$
3. $\sin^{-1} A + \sin^{-1} B = ?$
 - a. $\sin^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$
 - b. $\cos^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$
 - c. $\sin^{-1}(A\sqrt{1-B^2} - B\sqrt{1-A^2})$
 - d. None of these
4. Find the derivative of $\sin(\sin x)$ w.r.t. x
 - a. 0
 - b. $\cos(\cos x)$
 - c. $\cos(\sin x) \sin x$
 - d. $\cos(\sin x) \cos x$
5. $\int_{-1}^1 \ln e^x dx$
 - a. 0
 - b. 1
 - c. $e^2(\cos e^x - \sin e^x)$
 - d. $e^2(\cos e^x + \sin e^x)$
6. $\int e^x (\cos e^x - \sin e^x) dx$
 - a. $(\cos e^x + \sin e^x)$
 - b. $(\cos e^x - \sin e^x)$
 - c. $e^2(\cos e^x - \sin e^x)$
 - d. $e^2(\cos e^x + \sin e^x)$
7. $\frac{(x+4)^2}{(y-5)^2} = 1$ is a hyperbola, find its centre and vertex?
 - a. $(-4, 5); (7, 5); (11, 5)$
 - b. $(-4, 5); (-7, 5); (-11, 5)$
 - c. $(4, 5); (-7, 5); (-11, 5)$
 - d. $(-4, 5); (7, 5); (11, 5)$
8. Six coins are tossed simultaneously, find the probability of getting at least one head?
 - a. $\frac{1}{64}$
 - b. $\frac{1}{8}$
 - c. $\frac{1}{32}$
 - d. $\frac{63}{64}$
9. $\tan^{-1} x = ?$
 - a. $\tan^{-1} x$
 - b. $\tan^{-1} x$
 - c. $\tan^{-1} x$
 - d. None of these
10. A car has four seats, Akbar, Anjad, Aslam and Vickey are taking their seats, if Aslam is driving then how many ways the passenger can be arranged?
 - a. 16
 - b. 24
 - c. 6
 - d. 9
11. Find the equation of tangent from $(\frac{1}{3}, \frac{4}{9})$ to the hyperbola $\frac{(x)^2}{16} - \frac{(y)^2}{9} = 1$
 - a. $24xy - 6x + 63$
 - b. $9y = 27x - 3$
 - c. $27x - 9y - 5 = 0$
 - d. None of these
12. If z is a complex number then express $2x - 4y - 3$ in conjugate coordinate form
 - a. $(Z + \bar{Z}) - 3 - 2i(Z - \bar{Z})$
 - b. $(Z - \bar{Z}) - 3 = 2(Z - \bar{Z})$
 - c. $(Z + \bar{Z}) + 3 = 2(Z - \bar{Z})i$
 - d. $(Z + \bar{Z}) - 3 = 2(Z + \bar{Z})i$
13. Find the imaginary part of $z = (x + 2y)^2$
 - a. 0
 - b. $-4y^2$
 - c. $-4xy$
 - d. $x^2 - y^2$
14. Find the slope of line perpendicular to $5x - 3y + 1 = 0$
 - a. $-\frac{3}{5}$
 - b. $\frac{5}{3}$
 - c. $-\frac{5}{3}$
 - d. $\frac{3}{5}$
15. Find the equation of line passing through $(0, 0)$, $(0, 2)$ and $(0, 5)$?
 - a. $y = 0$
 - b. $x = 0$
 - c. $y = x + 7$
 - d. $x = y + 7$
16. Area of triangle can be calculated using:
 - a. $\frac{1}{2}$ base $\times$ altitude
 - b. $\frac{1}{2}$ (product of adjacent sides $\times$ sine of angle in between sides)
 - c. Hero's formula
 - d. All of these
17. If $f(x) = 2x + 5$ the $f^{-1}(3)$ will be?
 - a. -2
 - b. -1
 - c. 0
 - d. 2
18. $-\sin 3\alpha = ?$

ANLES HUSSAIN

- a. $3 \sin a$
b. $\sin 3(-a)$
19. If $A = \begin{bmatrix} 1 & 2 \\ -3 & 2y \end{bmatrix}$ is a singular matrix, then y will be -
a. -3
b. -1
c. 3
20. The equation having roots $\{2, 1/3\}$ is
a. $3x^2 = 7x + 2$
b. $3x^2 = 7x - 2$
21. The general term of sequence 1, 4, 16, 25, ... is
a. $5n - 1$
b. n^2
c. $7x^2 - 3x + 2$
d. None of these
22. If cube roots of unity are 1, w , w^2 then cube roots of $1 + 6i$ will be?
a. ± 8
b. $2, 2w, 2w^2$
c. $2n^2, n$
d. $n^2 + 1$
23. $\int (120x^2 + 10) dx$
a. $2x^3 + 3x^2 + C$
b. $3x^3 + 10x + C$
c. $4x^3 + 3x^2 + C$
d. All of these
24. $\vec{A} = (3, 2, 4)$ is perpendicular to $\vec{B} = (-3, a, -1)$, then value of a will be
a. $2/3$
b. $4/5$
c. $3/7$
d. $7/9$
25. Perimeter of a semi-circle when radius is equal to 6 cm
a. 36π
b. 6π
c. $16\pi + 12$
d. $6\pi + 12$



95. The reaction $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$ proceeds slower because of _____ activation energy of CO.
- A. equilibrium
 - B. constant
 - C. low
 - D✓ high
96. For the first time in 1911, Henry Moseley used X-rays for the determination of:
- A. Atomic number
 - B. Atomic mass
 - C. Molecular mass
 - D✓ Equivalent mass
97. The rate of a chemical reaction is directly proportional to the product of active masses of the reactants, it is referred to as:
- A. Law of conservation of energy
 - B. Law of mass action
 - C. Law of conservation of mass
 - D. Active mass law
98. The phenomenon in which certain elements emit invisible radiations is called:
- A✓ Spectroscopy
 - B. Radioactivity
 - C. Gravimetry
 - D. Chromatography
99. Real gases behave ideally at:
- A. Low temperature and low pressure
 - B. Moderate temperature and low pressure
 - C✓ High temperature and low pressure
 - D. High temperature and high pressure
100. Which of the following properties depends upon the amount of matter present in the system?
- A. Density
 - B✓ Gibb's free energy
 - C. Pressure
 - D. Temperature

PHYSICS

41. Equations of kinetics are valid only when the acceleration is:
- A. Increasing ~
 - B. Decreasing
 - C. Constant
 - D. Vary with time
42. Law of heat exchange is used to determine:
- A. Coefficient of linear expansion
 - B. Coefficient of volume expansion
 - C. Ideal gas constant
 - D. Specific heat of substance
43. A solar day is the time interval between two successive appearances of the sun overhead. The time that is referred to rotation of the earth about its axis is called:
- A✓ Universal Time
 - B. Length of the day
 - C. Time interval
 - D✓ Solar day
44. In the photo electric effect, a photon of energy $h\nu$ is incident on the metal surface. The energy required to eject the valance electron from the surface of the metal is expressed as:
- A. Wave nature of the light
 - B✓ Compton scattering
 - C. Melting point of the metal
 - D. Work function of the metal
45. When an electron and its anti-particle positron come close enough to each other, they completely convert into radiation energy in the form of photons, this process is called:
- A. Photo electric effect
 - B. Pair production
 - C. Annihilation of matter
 - D✓ Compton effect

6. The guests broke a dozen glass at the party.

- A. The guests
- B. ☒ Broke
- C. Glass
- D. At
- E. No error

Questions 7-8

Television is typical of many new scientific words which are deliberately invented from old Greek and Latin words. In this case the prefix 'tele' is Greek and means 'far' (of telephone, telegram), while the root 'vision' is derived from the Latin verb meaning 'to see'.

7. The word "Television" is invented from:

- A. English and Spanish
- B. French and Arabic
- C. Dutch and German
- D. ☒ Greek and Latin

8. Which of the following gives the meaning of the prefix "tele"?

- A. near
- B. close
- C. ☒ far
- D. away

Choose the lettered word or phrase that is most nearly opposite in meaning to the word in capital letters.

9. PATRIOTISM:

- A. nationalism
- B. socialism
- C. obsession
- D. ☒ disloyalty

10. SUPREMACY:

- A. excellency
- B. sovereignty
- C. ☒ inferiority
- D. influence

ENGLISH

Choose the word most similar in meaning to the capitalized one.

1. SHABBY:

- A. organized
- B. untidy
- C. reluctant
- D. sensible

2. FORBID:

- A. prohibit
- B. chide
- C. permit
- D. constraint

Complete the sentences by choosing the most appropriate option, from the given lettered choices (A to D/E) below each.

3. The road is wet. It must _____ raining.

- A. have been
- B. had
- C. has
- D. have

4. He would like _____ have a job in the same office as his brother.

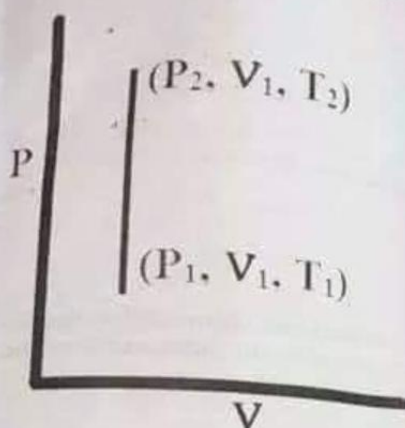
- A. too
- B. to
- C. of to
- D. get

Identify the word or phrase that needs to be changed for the sentence to be correct.

5. It's too late to go for a walk now; besides, it's beginning to rain.

- A. It's
- B. Go
- C. Besides
- D. Beginning
- E. No Error

57. Which process is shown in the graph between pressure and volume?



- A. Adiabatic
- B. Isobaric
- C. Isochoric
- D. ☒ Isothermal

58. If in a circuit, 2 ampere current is drawn from the battery in 10 minutes. How much charge will flow through the circuit in this time?

- A. 1200 Coulombs
- B. 600 Coulombs
- C. 500 Coulombs
- D. 20 Coulombs

59. If the sunlight is incident on the photocell, the number of electrons emitted from the surface of the cell increases with the:

- A. Increase of light frequency
- B. Increase of light intensity
- C. Decrease of light frequency
- D. Decrease of light intensity

60. The process of natural decay of some heavy nuclides is because of the _____ phenomenon.

- A. Emission spectra
- B. Nuclear fusion
- C. Nuclear fission
- D. Radioactivity

S-19-3442-JSMU-BLUE-281018

61. Let X be a radioactive element having N_0 nuclei. The time required to decay this element to one half of its initial number N_0 is known as:
- A. Law of radioactive decay
 - B. Decay constant
 - C. Half life
 - D. Decay Time
62. Bimetallic thermostat is an example of:
- A. Electric field
 - B. Thermal expansion
 - C. Heat engine
 - D. Isobaric process
63. A cyclist moving towards right with an acceleration of 4m/Sec^2 . At $t=0$, he has travelled 5m moving towards the right at 15m/Sec . What will be his position at $t=2$ seconds?
- A. 36
 - B. 38
 - C. 41
 - D. 43
64. In pure semiconductor materials, some impurities are added to change the properties of these semiconductors. This process is called:
- A. Crystallography
 - B. Rectification
 - C. Intrinsic defects
 - D. Doping
65. Two forces equal in magnitude but opposite in direction and not acting along the same line constitute a couple. The movement of this couple will be:
- A. Dependent on the location of the origin
 - B. Independent of the location of origin
 - C. Zero
 - D. Scalar product
66. If we increase the area of a cross section of a conductor having a length L, the resistance R of that conductor will be:
- A. Zero
 - B. Constant
 - C. Increasing
 - D. Decreasing

90. In electrochemical series, elements are arranged in order of their standard electrode potentials, the correct decreasing reactivity order for metals is:

- A. Gold, silver, magnesium, aluminum
- B. Mercury, calcium, sodium, magnesium
- C. Sodium, aluminum, lead, copper
- D. Potassium, silver, magnesium, aluminum

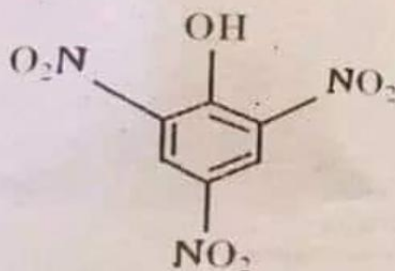
91. When electricity is passed through water in the presence of an acid oxygen _____ is produced.

- A. Carbon
- B. CO
- C. CO₂
- D. Hydrogen

92. The Neutron was discovered by:

- A. Goldstein
- B. Rutherford
- C. J.J Thomson
- D. Chadwick

93. The figure mentioned below is commonly known as:



- A. Resorcinol
- B. Aspirin
- C. Picric acid
- D. Mesityl

94. Which alkyl halide has the lowest reactivity for a particular alkyl group?

- A. R-Cl
- B. R-F
- C. R-Br
- D. R-I

- is:
46. During radioactive decay of a nucleus ${}_Z^AX^A$, β emission takes place along with daughter nucleus. Because of this beta particle emission, the mass number A of parent nucleus:
- A. Remains constant
 - B. Decreases by 1
 - C. Increases by 1
 - ☒ D. Decreases by 2
47. In simple harmonic motion, a particle is oscillating in such a way that its total energy remains conserved. When this particle reaches its two extreme positions, its potential energy at extreme positions becomes:
- A. Minimum
 - ☒ B. Maximum
 - C. Zero
 - D. Greater than its kinetic energy
48. In electronic circuits, a PN junction diode is used to convert alternating current into direct current. This process is called:
- ☒ A. Rectification
 - B. Amplification
 - C. Doping
 - D. Integrated circuit
- t on
49. The process in which light bends around an obstacle is called:
- ☒ A. Diffraction of light
 - B. Interference of light
 - C. Reflection of light
 - D. Polarization of light
50. The atomic number Z of an atom represents the:
- A. Number of protons
 - B. Number of electrons and protons
 - C. Number of neutrons
 - D. Sum of protons and neutrons
51. If a car collides with a housefly, what will be the magnitude of the force experienced by the housefly?
- A. Much greater than the car experienced by the housefly
 - B. Much lesser than the car experienced by the housefly
 - C. Same as the car experienced by the housefly
 - D. 10 times less than the car experienced by the housefly

quired
own

67. If constant temperature conditions are applied to a gas container, then the volume of the mass of that gas tends to increase by:

- A. Decreasing the applied pressure
- B. Increasing the applied pressure
- C. Increasing applied pressure with double force
- D. Increasing the weight

68. The efficiency of the Carnot engine working between temperatures 177°C and 77°C will be:

- A. 22.22 %
- B. 23.23 %
- C. 25.25 %
- D. 30.30 %

2.
Sec.

69. Let B be the magnetic field and a conductor of length L is moved across this field. Due to this activity, a potential difference appears across the ends of the conductor. This type of potential difference is termed as:

- A. Self-Induction
- B. Mutual Induction
- C. Motional E.M.F
- D. Electrostatic potential

70. Which of the following phenomenon represents the wave nature of light?

- A. Photoelectric effect
- B. Compton scattering
- C. Interference of light
- D. Pair production

84. How many significant figures are there in 00.4793?

- A. 3
- B. 4
- C. 5
- D. 6

85. Which one of the following is NOT a meta directing group?

- A. COOH
- B. SO_3H
- C. CHO
- D. OH

86. How many chain isomers are shown by C_5H_{12} ?

- A. 5
- B. 4
- C. 3
- D. 2

87. For the following exothermic reaction, decrease in temperature will shift the equilibrium towards: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$

- A. Left
- B. The point of equilibrium
- C. Both directions
- D. Right

88. In which of the following nitrogen compounds, N has the highest oxidation state?

- A. NH_2OH
- B. N_2H_4
- C. HNO_3
- D. NH_3

89. The boiling point of water is highest than other hydrides because water molecules can form:

- A. 4 hydrogen bonds
- B. 3 hydrogen bonds
- C. 2 hydrogen bonds
- D. 1 hydrogen bonds

77. Which of the following reactions is NOT shown by Ketones?
- A. Reaction with HCN
 - B. Reaction with Fehling solution-
 - C. Reaction with NaHSO_3
 - D. Reaction with 2,4-dinitrophenyl-hydrazine
78. Which one of the following is the pure carbon compound and used as reducing agent in industries?
- A. Coal tar✓
 - B. Petroleum
 - C. Coal gas
 - D. Coke
79. The alloy Dura lumin is composed of _____ of Al.
- A. 95%
 - B. 90%
 - C. 85%
 - D. 80%
80. The normal pH of blood is:
- A. 7.75
 - B✓ 7.35
 - C. 7.25
 - D. 7.05
81. The number of bond(s) between carbon and nitrogen atoms in a Nitrile is:
- A. One sigma and one pi
 - B. Two sigma and one pi
 - C. Only sigma
 - D. One sigma and two pi
82. Methyl orange is _____ in acidic solutions.
- A. yellow
 - B. pink
 - C. orange
 - D. red
83. The melting point of NaCl is very high 801°C , it is reduced to 600°C by addition of _____ in Down's process.
- A. Calcium chloride
 - B. Magnesium chloride
 - C. Aluminum chloride
 - D. Potassium chloride

CHEMISTRY

71. Which one of the following does NOT give iodoform test?
- A. CH_3OH
 - B. $\text{CH}_3\text{CH}_2\text{OH}$
 - C. CH_3CHO
 - D. CH_3COCH_3
72. 106 gram of Na_2CO_3 per dm^3 of solution of Na_2CO_3 in water, the concentration of the solution will be:
- A. 1N
 - B. 0.1M ✓
 - C. 1M
 - D. 0.02M
73. The transparent plastic used to make combs and hair brushes is called:
- A. PVA
 - B. PVC -
 - C. Bakelite
 - D. Perspex
74. Which one of the following vitamins is required for normal growth, vision and keeping the skin healthy?
- A ✓ Vitamin D
 - B. Vitamin E
 - C. Vitamin A
 - D. Vitamin K
75. Zwitter ion is formed when proton goes from:
- A. Amino to carboxyl group
 - B. Carboxyl to amino group
 - C ✓ Carboxyl group only
 - D. Amino group only
76. Lithium and beryllium are unique in such a way that they have higher charge densities which produce strong polarizing effects due to:
- A ✓ Nonmetal
 - B. Solubility
 - C. Small size
 - D. Large size

52. There are different types of waves that exist in this universe. Which of the following waves do NOT need any material medium to travel?

- A. Sound waves
- B. ✓ Waves produced on a string
- C. Electromagnetic waves
- D. Waves produced on the water surface

53. If a dielectric material is placed between two plates of a capacitor, the net capacitance of the capacitor:

- A. Decreases
- B. Increases
- C. Remains constant
- D. Zero

54. In free states, the mass of the nucleus is always:

- A. Greater than the mass of its constituents
- B. Equal to the mass of its constituents
- C. Less than the mass of its constituents
- D. Far greater than the mass of its constituents

55. If constant current is flowing through a coil then flux through that coil will become constant. In such type of arrangements, the electromotive force emf induced in coil will be:

- A. Self-inductance
- B. Zero
- C. Mutual-inductance
- D. Back emf

56. In electromagnetic spectrum of radiations, the wavelength spectrum that is visible to humans lies in the range of:

- A. 100-400 nm
- B. ✓ 400-700 nm
- C. 700-1000 nm
- D. 1000-1500 nm

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ADMISSION TEST # 2013

2013 DawoodTECHNOLOGY
TEST PAPER

UNIVERSITY OF ENGINEERING &

SECTION - I (ENGLISH)

Choose the word most similar in meaning to the capitalized one,

1. MONOTONOUS:

- (a) Limited (b) Excitable (c) Extremist (d) Repeatedly
(e) Conclusive

2. PRECISE:

- (a) Concise (b) Comprehensive (c) Exaggerate (d) Shortcoming
(e) Explanatory

3. DESTITUTE:

- (a) Disable (b) Isolated (c) Poor (d) Emangled
(e) enalize

4. EXPEDIENT

- (a) Scapegoat (b) Heterogeneous (c) Suitable (d) Apathetic
(e) Concentrate

Choose the lettered word or phrase that is most nearly opposite in meaning to the word in capital letters.

5. ORTHODOX: *→ ancient mingled*

- (a) Emangled (b) Unconventional (c) Conventional (d) Traditional
(e) Casual

6. ECONOMICAL:

- (a) Maddest (b) Efficient (c) Extravagant (d) Attentive
(e) Expensive

N.E.P. PAST PAPERS

	(a)	Modest	(b)	Artificial	(c)	Eschew	(d)	
		Attractive	(e)	Expansive				
7.	PLEASANT		(b)	Mercy	(c)	Comfort	(d)	Pleasure
	(a)	Happiness						
		Stylish						
8.	EXHIBIT		(b)	Showy	(c)	To act	(d)	Conserve
	(a)	Conceal	(b)	Reveal				
9.	BRITTLE		(b)	Noise	(c)	Tough	(d)	Wise
	(a)	Easy						
10.	EMISSARY		(b)	Spy	(c)	Merit	(d)	Achieve
	(a)	Director						
11.	JOYOUS		(b)	Sad	(c)	Gay	(d)	Morose
	(a)	Genial						
12.	PLEAD		(b)	Justify	(c)	Sue	(d)	pray
	(a)	Demand						
13.	VENERATE		(b)	Adapt	(c)	Correlate	(d)	Involve
	(a)	Abominate						

Identify the word or phrase that needs to be changed for the sentence to be correct:

14. The teacher gave Marta a failing grade on her composition because it was the same a composition he had already read. A B C D E
15. Your recipe for chicken is like to a recipe that my mother has. A B C D E
16. Jayne's apartment is very differs from Bill's even though they are in the same building. A B C D E
17. The less one earn, the lesser one must pay in income taxes. A B C D E

FOR FREE TEST PREPARATION CONTACT

Mohammad Amin Sharif [Aptitude test Expert]

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Read the passage to answer questions 7 – 8

Throughout history, the search for salt has played an important role in society. When it was scarce, salt was traded ounce for ounce with gold. Rome's major highway was called the Via Salaria, that is, the Salt Road. Along that road, Roman soldiers transported salt crystals from the salt flat at Onia up the Tiber River. In return, they received a salarium or salary, which was literally money

paid to soldiers to buy salt. The old saying "worth their salt", which means to be valuable, derives from the custom of payment during the Empire.

N.E.D. PAST PAPERS

18. What does the passage mainly discuss?
- (a) the old saying "worth their salt" (b) the Roman Empire
(c) salt (d) ancient trade (e) the Roman soldier
19. According to the passage, salt flats were located in:
- (a) Rome (b) Ostia (c) Customs (d) Tiber
(e) Salaria

Read the passage to answer questions 7 - 8

In May 1966, the World Health Organization was authorized to initiate a global campaign to eradicate smallpox. The goal was to eliminate the disease in one decade. Because similar projects for malaria and yellow fever had failed, few believed that smallpox could actually be eradicated, but eleven years after the initial organization of the campaign, no cases were reported in the field.

20. Which of the following is the best title for the passage?
- (a) The World Health Organization (b) The eradication of smallpox
(c) Smallpox vaccinations (d) Infectious diseases
(e) Vaccinations of entire villages
21. It can be inferred that:
- (a) No new cases of smallpox have been reported this year
(b) Malaria and yellow fever have been eliminated
(c) Smallpox victims no longer die when they contract the disease
(d) Smallpox is not transmitted from one person to another
(e) Extensive reporting of outbreak

Complete the sentences by choosing the most appropriate word, from the choices given below each.

22. Dennis Chavez of New Mexico _____ to the House of Representatives in 1930 and to the senate in 1938.
- (a) when elected (b) elected (c) who was elected (d) was elected
(e) was electing
23. Knee is the joint _____ the thigh bone meets the large bone of the lower leg.
- (a) where it is (b) where (c) why (d) when
(e) which
24. Quails typically have short rounded wings that enable _____ spring into full flight instantly when disturbed in _____ their hiding places.
- (a) they (b) to their (c) its (d) them to
(e) in

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25. When water freezes in the cracks of rocks _____ expands, causing the rocks to break.
 (a) apart (b) but (c) then (d) and
 (e) on

SECTION - II (MATHEMATICS) (For Pre-Engineering Group)

1. If $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$, then cofactor of 2nd row are
 (a) -3, 3, -11 (b) 3, -3, 11 (c) 3, 3, -11 (d) 3, 3, 11
2. If $2\Delta = a + b + c$, then the value of $\Delta = \begin{bmatrix} \Delta + c & a & b \\ c & \Delta + a & a \\ c & a & \Delta + b \end{bmatrix}$ is
 (a) $4\Delta^2$ (b) $2\Delta^3$ (c) Δ^4 (d) $3\Delta^2$
3. If $a, b \in \mathbb{R}$ then roots of $(x-a)(x-b) = abx^2$ are always
 (a) real (b) dependent on a (c) not real (d) dependent on a & b both
4. $12 \cdot 3456 =$
 (a) $12 \frac{3456}{9900}$ (b) $12 \frac{3422}{9900}$ (c) $12 \frac{122222}{9900}$ (d) B & C both
5. If a, b, c are in A.P., then $3^a, 3^b, 3^c$ are in
 (a) A.P. (b) G.P. (c) H.P. (d) none of these
6. Fifteen girls compete in a race. The first three places can be taken by them in _____ ways.
 (a) $12!$ (b) 45 (c) $15!$ (d) $15 \times 14 \times 13$
7. An experiment yields 3 mutually exclusive and exhaustive events A, B, C. If $P(A) = 2P(B) = 3P(C)$ then, $P(A) =$ _____
 (a) $\frac{1}{11}$ (b) $\frac{2}{11}$ (c) $\frac{3}{11}$ (d) $\frac{6}{11}$
8. If the 4th term in the expansion of $(px + x^{-1})^m$ is 2.5 for all $x \in \mathbb{R}$, then
 (a) $P = \frac{5}{2}, m = 3$ (b) $P = \frac{1}{2}, m = 6$ (c) $P = \frac{1}{2}, m = 4$ (d) none of these

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9. $1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 8^3 + 9^3 + 10^3 =$
 (a) 55 (b) 55^2 (c) 2682 (d) none of these
10. If the elevation of the sun is 30° , then the length of the shadow cast by a tower of 150 ft height is
 (a) $75\sqrt{3}$ ft (b) $\frac{150}{\sqrt{3}}$ ft (c) $150\sqrt{3}$ ft (d) none of these
11. The value of parameter 'a' for which the function $f(x) = 1 + ax$, $a \neq 0$ is the inverse of itself, is
 (a) -2 (b) -1 (c) 1 (d) 2
12. Limit of sequence $\frac{1 \cdot 2}{2 \cdot 3}, \frac{3 \cdot 4}{4 \cdot 5}, \frac{5 \cdot 6}{6 \cdot 7}, \dots$
 (a) 0 (b) 1 (c) ∞ (d) undefined
13. $\lim_{x \rightarrow \infty} \frac{4x^3 + 3x^2 - 2x}{1 - 2x + 3x^2 + 4x^3} =$
 (a) 4 (b) 0 (c) 1 (d) -1
14. $\lim_{x \rightarrow 0} \frac{\ln(1 + 2x)}{3x} =$
 (a) does not exist (b) undefined (c) 2 (d) $\frac{2}{3}$
15. The derivation of function $f(x) = \sin^{-1} \sqrt{1 - x^2}$ w.r.t $g(x) = \cos^{-1}(2x^2 - 1)$ is
 (a) $\tan^{-1}(1 + 2x)$ (b) $\cos^{-1}(2x + 1)$ (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$
16. The velocity of a particle moving along a straight line is given by $x = 3t + t^2$. The acceleration of the particle after 4 seconds from the start is
 (a) $\frac{d^2 v}{dt^2}$ (b) 2 (c) 11 (d) 11 units / sec²
17. The integral that can be evaluated is
 (a) $\int_0^1 \frac{dx}{x-1}$ (b) $\int_0^{\frac{\pi}{2}} \sin x \, dx$ (c) $\int_1^2 \sqrt{1-x^2} \, dx$ (d) $\int_{-2}^1 \ln x \, dx$

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18. The equation of the chord of the circle $x^2 + y^2 - 4x = 0$ where mid point is (1,0) is _____
 (a) $y = 2$ (b) $y = 1$ (c) $x = 2$ (d) $x = 1$
19. The slope of the normal at the point $(at^2, 2at)$ of the parabola: $y^2 = 4ax$ is _____
 (a) $\frac{1}{t}$ (b) t (c) $-t$ (d) $-\frac{1}{t}$
20. If $x + y + 1 = 0$ touches the parabola: $y^2 = \lambda x$, then λ is _____
 (a) $\frac{2}{3}$ (b) 4 (c) 6 (d) 8
21. The vector $2\vec{i} + 4\vec{j} + 7\vec{k}$ is perpendicular to the vector $2\vec{i} + 6\vec{j} + x\vec{k}$ when $x =$ _____
 (a) 0 (b) 2 (c) 4 (d) 7
22. If $w^{123} + w^{23} + w^{13} - 1 = 0$ then the least natural number n is _____
 (a) 0 (b) 4 (c) 6 (d) 2
23. The measure of the two sides of a triangle of area 6 units² are 4 & 5 cms. Its third side is _____
 (a) 3 (b) 3 units (c) 3 cms (d) none of these
24. $\int \sin \sqrt{2x} \, dx =$ _____
 (a) $\sqrt{2x} \cos \sqrt{2x} + \sin \sqrt{2x}$ (b) $-\sqrt{2x} \cos \sqrt{2x} + \sin \sqrt{2x} + C$
 (c) $-\sqrt{2x} \sin \sqrt{2x} + \cos \sqrt{2x} + C$ (d) None of these
25. The volume of the parallelepiped whose three adjacent edges are represented by the vectors :
 $\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$, $\vec{b} = \vec{i} - 2\vec{j} + \vec{k}$ & $\vec{c} = 3\vec{i} - \vec{j} + 2\vec{k}$, is _____
 (a) -7 (b) $\frac{7}{2}$ (c) $\frac{7}{3}$ (d) $\frac{7}{6}$

OR

SECTION - II (BIOLOGY) (For Pre-Medical Group)

1. Most peculiar cells of hydra are:

SECTION - III (PHYSICS)

1. A train moves on an horizontal surface with a constant velocity of 20 m/s. It requires maximum of power, the force retarding the motion is:
- (a) 1500 dyne (b) 15000 N (c) 515 dyne (d) 375 N
(e) 200N
2. Body sitting in train moving at a constant velocity throws a ball straight up into the air the ball will drop:
- (a) Behind him (b) In front of him (c) In his hand (d) Beside him
(e) Top of head
3. Newton's first law of motion is applicable to:
- (a) Inertial frames of reference (b) Relativistic frames of reference
(c) All frames of reference (d) International frame
(e) Smaller frames of reference
4. A spaceship orbiting the earth is an example of Newton's
- (a) 1st law (b) 2nd law (c) 3rd law (d) law of gravitation
(e) Coulomb's law
5. When a force 4N on a mass of 20kg for a time of 2s. what is the rate of change of momentum?
- (a) 1 kgms⁻² (b) 2kgms⁻² (c) 4kgms⁻² (d) 8kgms⁻²
(e) 10k gms⁻²
6. The S.I. unit of current is:

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- | | (a) Volt | (b) Candela | (c) Ampere | (d) Ohm | |
|--|--|--------------------------|-------------------|---------------------|---------------|
| 7. Henry is unit for | (a) Only mutual inductance | (b) Only self-inductance | (c) Both | | |
| self & mutual inductance | (d) Magnetic flux | (e) Relative inductance | | | |
| 8. Decibel is the unit of: | (a) Absorbing power (Absorbing co-efficient) | (b) Intensity of sound | (c) | | |
| Loudness of sound | (d) Frequency of sound | (e) | | | |
| Velocity of sound | | | | | |
| 9. 10^{-9} seconds are equivalent to: | (a) Deci seconds | (b) Nano seconds | (c) Milli seconds | | |
| (d) None of these | (e) Pico second | | | | |
| 10. Which of the following is biggest unit of energy? | (a) Joule | (b) Microjoule | (c) Electron volt | (d) None of these | |
| (e) B.T.U. | | | | | |
| 11. One Angstrom is equal to | (a) 10^{-8} cm | (b) 10^{-8} m | (c) 10^{-6} cm | (d) 10^8 mm | (e) |
| | 10^{-10} cm | | | | |
| 12. 1 Giga Hertz is equivalent to | (a) 10^{10} Hz | (b) 10^9 Hz | (c) 10^7 Hz | (d) 10^{100} Hz | (e) 10^{11} |
| Hz | | | | | |
| 13. The dimensional formula for acceleration is | (a) ML^2/T | (b) ML^2/T^2 | (c) MLT | (d) LT^2 | |
| (e) LT^3 | | | | | |
| 14. A vector in a given direction whose magnitude is unity is called | (a) Unit vector | (b) Parallel vector | (c) Free vector | (d) Position vector | |
| (e) Null vector | | | | | |
| 15. The x-component of the vector A making an angle of 30° with the y-axis is | (a) $A/\sqrt{2}$ | (b) $\sqrt{3}/2$ | (c) $0.5 A$ | (d) 0 | |
| (e) $6 A$ | | | | | |
| 16. If the resultant of two forces each of magnitude F is $2F$ then the angle between them will be | (a) 180° | (b) 30° | (c) 60° | (d) 0° | |
| (e) $3/2$ | | | | | |

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17. Two bodies are moving in opposite direction with speed v . What is the magnitude of their relative velocity with each other?
- (a) 0 (b) v (c) $\frac{v}{2}$ (d) $2v$
(e) $\frac{v^2}{2}$
18. Two forces of magnitude 20N each are acting at 30° & 60° with the x axis the y-component of the resultant force is approx:
- (a) 26.32N (b) 10N (c) 20×0.866 N (d) 15N
(e) 10×0.5 N
19. The center of gravity may be defined as a point at which
- (a) Gravitational force acts (b) The weight of the body acts
(c) The body acceleration due to gravity (d) The body stops its motion
(e) Nuclear force acts
20. The point at which whole weight of a body is concentrated is called
- (a) Centre of mass (b) Centre of gravity (c) Origin
(d) Centre of action (e) Point of rotation 2
21. The centre of gravity of a body of irregular shape lies
- (a) At its centre (b) At the intersection of medians (c) At the intersection
(d) At the surface of the body (e) At the interaction of vertex
22. A body or object is said to be in equilibrium when it
- (a) Moves with a variable velocity (b) Moves with a uniform velocity (c) Moves very fast in space
(d) Moves very slow in space (e) move with C (speed of light)
23. A body will be in translation equilibrium if the vectors sum of external force acting on a body is
- (a) Maximum (b) Minimum (c) Square (d) Zero
(e) Infinite
24. A bird sits on telegraph wire 3 km long. The additional tension produced in the wire is resultant
- (a) Infinite (b) Equal to the weight of the bird
(c) Less than the weight of the bird (d) More than the weight of the bird
(e) Zero
25. The co-efficient of frictional force between two surfaces in contact does NOT depend upon
- (a) The normal force passing one against the other (b) The area of surface
(c) Whether the surface are stationary or relative motion (d) Whether a lubricant is used or not
(e) When lubricant is heated

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SECTION - IV (CHEMISTRY)

- Heat provided to a system at constant pressure will be equal to
 (a) E (b) $P \times V$ (c) H (d) P
 (e) $E \times V$
- The temperature 23 K corresponds to
 (a) 0°C (b) 273°C (c) 100°C (d) 50°C
 (e) 20°C
- Volume of given mass of gas varies _____ with inverse of the temperature at constant pressure.
 (a) Inversely (b) Directly (c) Three times (d) Four times
 (e) Readily
- The number of atoms in 1 molecules in $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.
 (a) 36 (b) 72 (c) 108 (d) 197
 (e) 98
- Which one of the following atoms A, B, C, D or E would readily form an ion with a charge of 2?

	Mass No.	Atomic No.
(a)	12	6
(b)	16	8
(c)	24	12
(d)	31	15
(e)	30	15
- Which of the following is most likely to be the melting point of an ionic solid?
 (a) -180°C (b) -78°C (c) 114°C (d) 943°C
 (e) 900°C
- What volume does 400 cm^3 of sample of gas at 700 torr occupy when the pressure is changed to 1 atm.
 (assume temperature is constant)
 (a) 368 cm^3 (b) 184 cm^3 (c) 170 cm^3 (d) 870 cm^3 (e) 970 cm^3
- Dura - lumins is all of the following except
 (a) Used in making trains (b) It is composed of 95% Al + 4% Cu + 0.5% of Ni
 (c) Used in making aeroplanes (d) It is resistant to corrosion
 (e) It is oxides zing agent
- Aluminum Bronze contains
 (a) 90% Cu + 10% Al + 10% Al + 80% Mg (b) 90% Al + 10% Cu (c) 10% Cu + 90% Al

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- (d) 10% Mg + 10% Cu + 80% Al (e) 5% Mg + 10% Cu + 5% Al
10. Percentage of carbon in butanol is
(a) 50% (b) 55% (c) 60% (d) 65%
(e) 80%
11. The temperature at which a solid change into liquid is called its
(a) Critical point (b) Melting point (c) Boiling point (d) None of these
(e) Power point
12. Same solid when heated not melt but directly change into gaseous state it is
(a) Evaporation (b) Crystallization (c) Sublimation (d)
(e) Condensation (e) None
13. An element X forms compound having formula Na_2X_3 , XO_2 and OX_3 . What is X likely to be
(a) Carbon (b) Chlorine (c) Nitrogen (d) Sulphur
(e) Iodine
14. Which one of the following is the molecular formula of hydrocarbon which contains 80% by mass of carbon?
(a) CH_4 (b) C_2H_4 (c) C_2H_6 (d) C_3H_6
(e) C_4H_2
15. Elements which have completely filled outermost shell and do not combine with other elements are called
(a) Inert elements (b) Reactive elements (c) Unstable elements
(d) Rare earth elements (e) Typical
16. What is the conc of I_2 molecule in a solution containing 2.54 g of iodine in 250 cm^3 of solution?
(a) 0.01 M (b) 0.02 M (c) 0.04 M (d) 0.08 M
(e) 0.2 M
17. The formula for ordinary Potash - Alum (Phitkari) is
(a) $\text{K}_2\text{SO}_4\text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (b) $\text{K}_2\text{SO}_4\text{Cr}(\text{SO}_4)_3$
(c) $\text{K}_2\text{SO}_4\text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (d) $\text{K}_2\text{SO}_4\text{Al}_2(\text{SO}_4)_4 \cdot 12\text{H}_2\text{O}$
(e) K_2SO_4
18. Nitric Acid as all the following properties except
(a) it has acidic property (b) it has oxidizing property
(c) It has nitrating property (d) it as bleaching property
(e) it has reducing property
19. Highest oxidation state of Nitrogen is
(a) +5 (b) +4 (c) -3 (d) +2
(e) +6

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20. Alkenes are characterized by the presence of _____ bond between two carbon atoms.
(a) single (b) double (c) triple (d) none of the above
21. Alkenes are also called _____
(a) Paraffin (b) Olefins (c) Aldehydes (d) Ketones
(e) Carboxylic Acid
22. The correct name of (IUPAC): $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_3$
(a) Butane (b) 2-methyl propane (c) Isopropyle (d) Dimethyl ethane (e) None of these
23. An alkane hydrocarbon chain cyclization would result in the formation of _____
(a) Alicyclic compound (b) Aromatic compound (c) Ethene
(d) Cyclic compound (e) Esters
24. In 1913 Mosely found that _____ is the fundamental property of an atom.
(a) At. Number (b) At. Mass (c) At. Volume
(d) At. Charge (e) At. Structure
25. The amount of the energy required to remove an electron from an isolated atom is called?
(a) Ionization potential (b) Ionization ability (c) Electron affinity
(d) Electropositivity (e) Metallic character